Coding Assignment: Off-Policy Monte Carlo Prediction Using Importance Sampling

Course Information

- Course: Reinforcement Learning
- Topic: Off-Policy Monte Carlo Prediction Using Importance Sampling
- Language: Python 3
- Difficulty: Intermediate
- Estimated Time: 2–3 hours

Learning Objectives

After completing this assignment, students will be able to:

- Implement off-policy Monte Carlo prediction.
- Understand the difference between behavior and target policies.
- Implement ordinary importance sampling.
- Implement weighted importance sampling.
- Compare bias and variance of the two estimators.
- Analyze convergence properties.

Background

Suppose an agent interacts with an environment using a **behavior policy** b.

However, we wish to estimate the value function of another policy called the **target policy** π.

Since the collected episodes were not generated by π, we must correct the returns using **importance sampling**.

For an episode

$$S_0, A_0, R_1, \ldots, S_T$$

the importance sampling ratio is

$$\rho = \prod_{t=0}^{T-1} \frac{\pi(A_t|S_t)}{b(A_t|S_t)}$$

The ordinary importance sampling estimate is

$$V(s) = \frac{1}{N} \sum_{i=1}^{N} \rho_i G_i$$

The weighted importance sampling estimate is

$$V(s) = \frac{\sum_i \rho_i G_i}{\sum_i \rho_i}$$

Environment

Consider the following MDP.

States

A
B
Terminal

Actions

Left
Right

Transitions

State A

Action	Next State	Reward
Left	B	0
Right	Terminal	1

State B

Action	Next State	Reward
Left	Terminal	2
Right	Terminal	0

Behavior Policy

The behavior policy chooses actions uniformly.

P(Left)=0.5
P(Right)=0.5

Target Policy

The target policy always chooses

Left

Starter Code is attached

Tasks

Part 1

Generate **10,000** episodes.

Part 2

For every episode compute

- total return

$$G$$

Part 3

Compute the importance sampling ratio

$$\rho = \prod \frac{\pi(a|s)}{b(a|s)}$$

Hint:

rho = 1.0

for state, action, reward in episode:

```
    rho *= (
        target_policy[state][action]
        /
        behavior_policy[state][action]
    )
```

Part 4

Implement Ordinary Importance Sampling

Estimate

$$V(A)$$

using

$$V = \frac{1}{N} \sum \rho G$$

Part 5

Implement Weighted Importance Sampling

Estimate

$$V(A) = \frac{\sum \rho G}{\sum \rho}$$

Part 6

Repeat the experiment 20 times.

Store the estimated value each time.

Compute

- mean
- standard deviation

for

- ordinary IS
- weighted IS

Part 7

Plot

- Ordinary IS estimate vs experiment number
- Weighted IS estimate vs experiment number

Part 8

Answer the following questions.

Q1

Why is importance sampling necessary?

Q2

Why can the importance sampling ratio become very large?

Q3

Which estimator has smaller variance?

Q4

Which estimator is unbiased?

Q5

Why might weighted importance sampling converge faster in practice?

Deliverables

Submit

- Python source code
- plots
- written answers

Grading Rubric

Component	Points
Episode generation	15
Return computation	10
Importance ratio calculation	20
Ordinary importance sampling	15
Weighted importance sampling	15
Statistical analysis (mean and standard deviation)	10
Plots	5
Conceptual questions	10
Total	100